

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} 0 & M_{xy} & 0 \\ M_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xy} = \mathbb{M}_{xy}^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & M_{xy} & 0 \\ -M_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xy} = \mathbb{T}_z^{(1)}$$

* Rank 3 tensor (ss). * Rank 3 tensor (sa). * Rank 3 tensor (a). * Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & M_{yyxy} \\ 0 & 0 & 0 & 0 & 0 & M_{zzxy} \\ 0 & 0 & 0 & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & M_{yzzx} & 0 & 0 \\ M_{xxxy} & M_{yyxy} & M_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxy} = \mathbb{M}_{4z}^{\alpha(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} - \mathbb{M}_z^{\beta(1)}$$

$$M_{yyxy} = -\mathbb{M}_{4z}^{\alpha(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} - \mathbb{M}_z^{\beta(1)}$$

$$M_{zzxy} = \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_z^{\beta(1)}$$

$$M_{yzzx} = \mathbb{M}_{xy}^{(2)} + 2\mathbb{M}_z^{\beta(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & M_{yyxy} \\ 0 & 0 & 0 & 0 & 0 & M_{zzxy} \\ 0 & 0 & 0 & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & -M_{yzzx} & 0 & 0 \\ -M_{xxxy} & -M_{yyxy} & -M_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxy} = \mathbb{M}_{xy}^{(3)} + 2\mathbb{T}_z^{(1)} - 2\mathbb{T}_z^{\alpha(1)} + 2\mathbb{T}_z^{\beta(1)}$$

$$M_{yyxy} = \mathbb{M}_{xy}^{(3)} - 2\mathbb{T}_z^{(1)} + 2\mathbb{T}_z^{\alpha(1)} + 2\mathbb{T}_z^{\beta(1)}$$

$$M_{zzxy} = \mathbb{M}_{xy}^{(3)} - 4\mathbb{T}_z^{\beta(1)}$$

$$M_{yzzx} = -\mathbb{T}_z^{(1)} - 4\mathbb{T}_z^{\alpha(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & M_{yzzx} & 0 \\ M_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = 2\mathbb{M}_{xy}^{(4)}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & M_{yzzx} & 0 \\ -M_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = -\mathbb{T}_z^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & M_{xxxy} \\ 0 & 0 & M_{yyxy} \\ 0 & 0 & M_{zzxy} \\ 0 & M_{yzzx} & 0 \\ M_{zxyz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxy} = -2\mathbb{M}_{xy}^{(5)} + \mathbb{T}_z^{(3)} - \mathbb{T}_z^{\alpha(2)} + \mathbb{T}_z^{\beta(2)}$$

$$M_{yyxy} = 2\mathbb{M}_{xy}^{(5)} + \mathbb{T}_z^{(3)} - \mathbb{T}_z^{\alpha(2)} - \mathbb{T}_z^{\beta(2)}$$

$$M_{zzxy} = \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{(4)} + 2\mathbb{T}_z^{\alpha(2)}$$

$$M_{yzzx} = -\mathbb{M}_{xy}^{(5)} + \mathbb{T}_z^{(4)} - \mathbb{T}_z^{\alpha(2)} - \mathbb{T}_z^{\beta(2)}$$

$$M_{zxyz} = \mathbb{M}_{xy}^{(5)} + \mathbb{T}_z^{(4)} - \mathbb{T}_z^{\alpha(2)} + \mathbb{T}_z^{\beta(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & M_{zxyz} & 0 & 0 \\ M_{xyxx} & M_{xyyy} & M_{xyzx} & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = \mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} + \mathbb{T}_z^{\beta(3)}$$

$$M_{zxyz} = -\mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} - \mathbb{T}_z^{\beta(3)}$$

$$M_{xyxx} = -2\mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)} + \mathbb{T}_z^{\beta(3)}$$

$$M_{xyyy} = 2\mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)} - \mathbb{T}_z^{\beta(3)}$$

$$M_{xyzx} = \mathbb{T}_z^{(5)} + 2\mathbb{T}_z^{(6)} + 2\mathbb{T}_z^{\alpha(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{xxxy} & 0 & 0 & M_{xxyx} \\ 0 & 0 & 0 & 0 & 0 & M_{yyxy} & 0 & 0 & M_{yyyx} \\ 0 & 0 & 0 & 0 & 0 & M_{zzxy} & 0 & 0 & M_{zzyx} \\ 0 & 0 & 0 & 0 & M_{yzzx} & 0 & 0 & M_{yzxz} & 0 \\ 0 & 0 & 0 & M_{zxyz} & 0 & 0 & M_{zxzy} & 0 & 0 \\ M_{xyxx} & M_{xyyy} & M_{xyzx} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxy} = \mathbb{M}_{4z}^{\alpha(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)} - \mathbb{M}_z^{\beta(1)} + 2\mathbb{T}_z^{(1)} + \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} + 2\mathbb{T}_z^{\beta(1)} + \mathbb{T}_z^{\beta(2)}$$

$$M_{xxyx} = \mathbb{M}_{4z}^{\alpha(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_{xy}^{(5)} - \mathbb{M}_z^{\beta(1)} + 2\mathbb{T}_z^{(1)} - \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)} + 2\mathbb{T}_z^{\beta(1)} - \mathbb{T}_z^{\beta(2)}$$

$$M_{yyxy} = -\mathbb{M}_{4z}^{\alpha(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_{xy}^{(5)} - \mathbb{M}_z^{\beta(1)} - 2\mathbb{T}_z^{(1)} + \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} + 2\mathbb{T}_z^{\beta(1)} - \mathbb{T}_z^{\beta(2)}$$

$$M_{yyyx} = -\mathbb{M}_{4z}^{\alpha(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)} - \mathbb{M}_z^{\beta(1)} - 2\mathbb{T}_z^{(1)} - \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)} + 2\mathbb{T}_z^{\beta(1)} + \mathbb{T}_z^{\beta(2)}$$

$$M_{zzxy} = \mathbb{M}_{xy}^{(1)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_z^{\beta(1)} + \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{(4)} + 2\mathbb{T}_z^{\alpha(2)} - 4\mathbb{T}_z^{\beta(1)}$$

$$M_{zzyx} = \mathbb{M}_{xy}^{(1)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_z^{\beta(1)} - \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{(4)} - 2\mathbb{T}_z^{\alpha(2)} - 4\mathbb{T}_z^{\beta(1)}$$

$$M_{yzzx} = \mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(5)} + 2\mathbb{M}_z^{\beta(1)} - \mathbb{T}_z^{(1)} + \mathbb{T}_z^{(4)} - 4\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} - \mathbb{T}_z^{\beta(2)}$$

$$M_{yzxz} = \mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(5)} + 2\mathbb{M}_z^{\beta(1)} - \mathbb{T}_z^{(1)} - \mathbb{T}_z^{(4)} - 4\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)} + \mathbb{T}_z^{\beta(2)}$$

$$M_{zxyz} = \mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(5)} + 2\mathbb{M}_z^{\beta(1)} + \mathbb{T}_z^{(1)} + \mathbb{T}_z^{(4)} + 4\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} + \mathbb{T}_z^{\beta(2)}$$

$$M_{zxzy} = \mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(5)} + 2\mathbb{M}_z^{\beta(1)} + \mathbb{T}_z^{(1)} - \mathbb{T}_z^{(4)} + 4\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)} - \mathbb{T}_z^{\beta(2)}$$

$$M_{xyxx} = \mathbb{M}_{4z}^{\alpha(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(3)} - \mathbb{M}_z^{\beta(1)} - 2\mathbb{T}_z^{(1)} + 2\mathbb{T}_z^{\alpha(1)} - 2\mathbb{T}_z^{\beta(1)}$$

$$M_{xyyy} = -\mathbb{M}_{4z}^{\alpha(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(3)} - \mathbb{M}_z^{\beta(1)} + 2\mathbb{T}_z^{(1)} - 2\mathbb{T}_z^{\alpha(1)} - 2\mathbb{T}_z^{\beta(1)}$$

$$M_{xyzx} = \mathbb{M}_{xy}^{(1)} - \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_z^{\beta(1)} + 4\mathbb{T}_z^{\beta(1)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{yzzx} & 0 & 0 & M_{yzxz} & 0 \\ 0 & 0 & 0 & M_{zxyz} & 0 & 0 & M_{zxzy} & 0 & 0 \\ M_{xyxx} & M_{xyyy} & M_{xyzx} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} M_{yzzx} &= 2\mathbb{M}_{xy}^{(4)} + \mathbb{M}_{xy}^{(6)} - \mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} + \mathbb{T}_z^{\beta(3)} \\ M_{yzxz} &= -2\mathbb{M}_{xy}^{(4)} + \mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} + \mathbb{T}_z^{\beta(3)} \\ M_{zxyz} &= 2\mathbb{M}_{xy}^{(4)} - \mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} - \mathbb{T}_z^{\beta(3)} \\ M_{zxzy} &= -2\mathbb{M}_{xy}^{(4)} - \mathbb{M}_{xy}^{(6)} - \mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} - \mathbb{T}_z^{\beta(3)} \\ M_{xyxx} &= -2\mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)} + \mathbb{T}_z^{\beta(3)} \\ M_{xyyy} &= 2\mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)} - \mathbb{T}_z^{\beta(3)} \\ M_{xyzx} &= \mathbb{T}_z^{(5)} + 2\mathbb{T}_z^{(6)} + 2\mathbb{T}_z^{\alpha(3)} \end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & M_{xxxy} & & 0 \\ M_{yyyyx} & 0 & & 0 \\ 0 & 0 & & 0 \\ 0 & 0 & & 0 \\ 0 & M_{zzxy} & & 0 \\ M_{xxyx} & 0 & & 0 \\ M_{yzzx} & 0 & & 0 \\ 0 & 0 & & 0 \\ 0 & M_{xyyy} & & 0 \\ 0 & 0 & \frac{M_{xxxy}}{2} - \frac{M_{xxyx}}{2} + M_{xyyy} - M_{yyyyx} + M_{zzxy} & \end{bmatrix}$$

$$\begin{aligned} M_{xxxy} &= \mathbb{M}_{4z}^{\alpha(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)} - \mathbb{M}_z^{\beta(1)} \\ M_{yyyyx} &= -\mathbb{M}_{4z}^{\alpha(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)} - \mathbb{M}_z^{\beta(1)} \\ M_{zzxy} &= \mathbb{M}_{xy}^{(1)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_z^{\beta(1)} \\ M_{xxyx} &= \mathbb{M}_{4z}^{\alpha(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_{xy}^{(5)} - \mathbb{M}_z^{\beta(1)} \\ M_{yzzx} &= \mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(5)} + 2\mathbb{M}_z^{\beta(1)} \\ M_{xyyy} &= -\mathbb{M}_{4z}^{\alpha(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(3)} - \mathbb{M}_z^{\beta(1)} \end{aligned}$$